Personal Statement

MSc in Computer Science | City University of Hong Kong (Dongguan)

The robots I have worked on have often exposed their first serious problems in software rather than in the control law. On the multi-robot platform I develop for at Kaihong, motion-control code could not be reused across hardware until timestamps, coordinate frames and validity checks had been standardised across sensors and actuators; inconsistent state information leads directly to unreliable control. This experience clarified the direction I want to pursue: developing the software, data and distributed-system foundations that allow control and AI algorithms to operate reliably across robot platforms. I have been learning this layer through practical work rather than systematic study, and the MSc in Computer Science is where I intend to build those foundations deliberately.

I am a final-year Computer Science student at Shenzhen Technology University, ranked 6 of 156, with Principles of Automatic Control and Modern Control Theory taken alongside the core curriculum. My computing background has been tested most directly on Kaihong's M-Robots platform on OpenHarmony. I built a unified capability model for sensors, actuators and controllers in C++ and mapped ROS and ROS2 motion commands onto low-level drivers; implemented device discovery, capability registration, state synchronisation and task dispatch over M-DDS; and gated task plans through an action-proposal, policy-check, execution and state-feedback chain. These tasks required interface design, distributed coordination and execution safety, and showed me why platform architecture cannot be learned only through ad hoc project work.

The rest of my work has been control and navigation implemented as software on real hardware. At the Shenzhen Institute of Artificial Intelligence and Robotics for Society I was responsible for the balance control of a dual-arm ballbot, from a deployed PID cascade and LQR analysis of the system as a four-state inverted pendulum to the ROS-to-CAN execution chain and fall-recovery safeguards. In laboratory projects I implemented gain-scheduled LQR and virtual model control on STM32H7/FreeRTOS so that a wheel-legged robot stays balanced as its leg length changes, and the complete EKF/AMCL, Hybrid-A*, TEB and MPC chain that lets a car-like vehicle localise, plan around obstacles and follow its path. In the university RoboMaster team I owned the gimbal, launcher and chassis control loops of the hero robot that reached the 2025 national final. Across these projects, I had to manage real-time constraints, inconsistent timing and coordinate frames in multi-source data, and deliver a complete, documented system rather than an isolated module.

My AI experience is more limited in scale but covers a complete application: an OpenHarmony smart gardening system I built alone, with YOLOv5 inference quantised to INT8 on an RK3588 edge device, a LangGraph agent chain for decisions and a Hi3861 irrigation state machine. It showed me how to place a model within a working product loop; it also showed me that larger fleets require stronger data engineering, evaluation methods and security design than I have formally studied.

The programme's stated aims align closely with the gaps I have identified. The first, to strengthen the software development capability of computer professionals, addresses the capability I have been developing at Kaihong. The second, to deepen understanding of artificial intelligence, data science and information security, points directly to gaps I can identify in my work: a fleet whose tasks are dispatched over a distributed bus needs a security model as well as a scheduler, and I have not yet received systematic training in designing one. The learning outcomes, applying network, software and data engineering concepts to design quality software and identifying key issues in a specific area, match what I want to do for robot platforms through established methods rather than ad hoc solutions. The two-year structure would let me study the foundations first and then devote a focused project to platform software or learning-based perception for mobile robots. Dongguan sits inside the Greater Bay Area's robotics and manufacturing belt, where platform software of this kind is being built and deployed.

After the degree I intend to work in the Greater Bay Area as a software engineer for robot platforms and motion algorithms, and in time to lead the software side of a robot product. A Computer Science degree is a broader route to that goal than a robotics-specific one, and I am choosing it deliberately: software engineering principles transfer across robot platforms, while this is the area in which my training is least systematic relative to how much I rely on it. My previous projects have taken control and platform modules from software design to working hardware, and have also shown me the software foundations I still need to build them properly.